# William Stallings Computer Organization and Architecture 10th Edition

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# Chapter 14

## Processor Structure and Function

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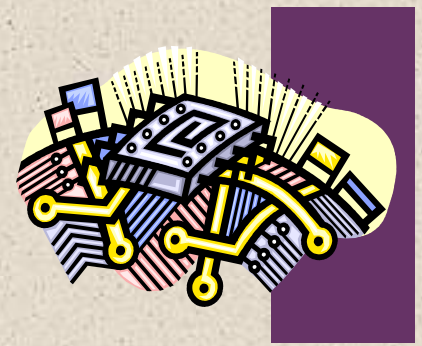
# Processor Organization



## Processor Requirements:

- Fetch instruction
  - The processor reads an instruction from memory (register, cache, main memory)
- Interpret instruction
  - The instruction is decoded to determine what action is required
- Fetch data
  - The execution of an instruction may require reading data from memory or an I/O module
- Process data
  - The execution of an instruction may require performing some arithmetic or logical operation on data
- Write data
  - The results of an execution may require writing data to memory or an I/O module
- In order to do these things the processor needs to store some data temporarily and therefore needs a small internal memory

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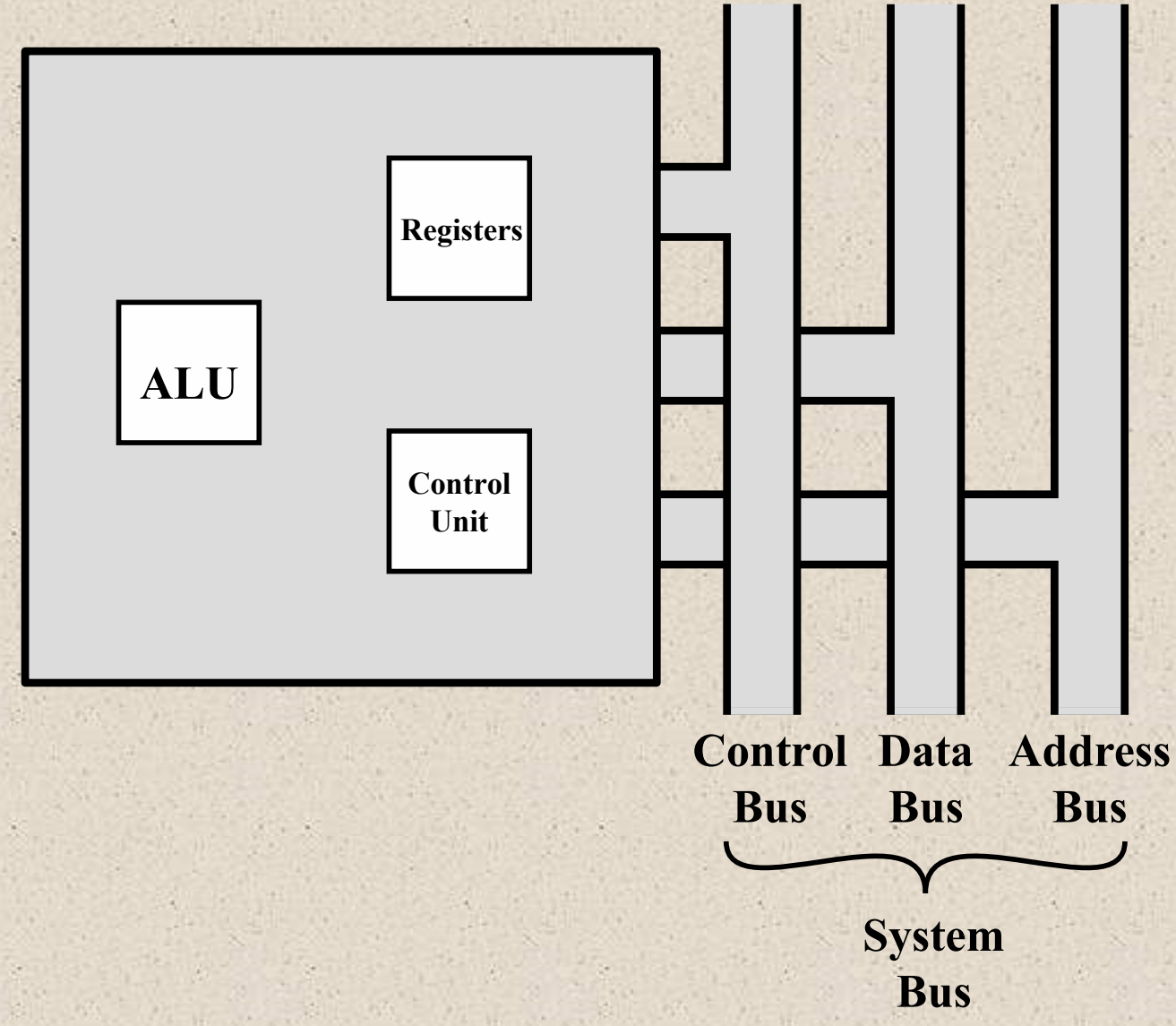


**Figure 14.1 The CPU with the System Bus**

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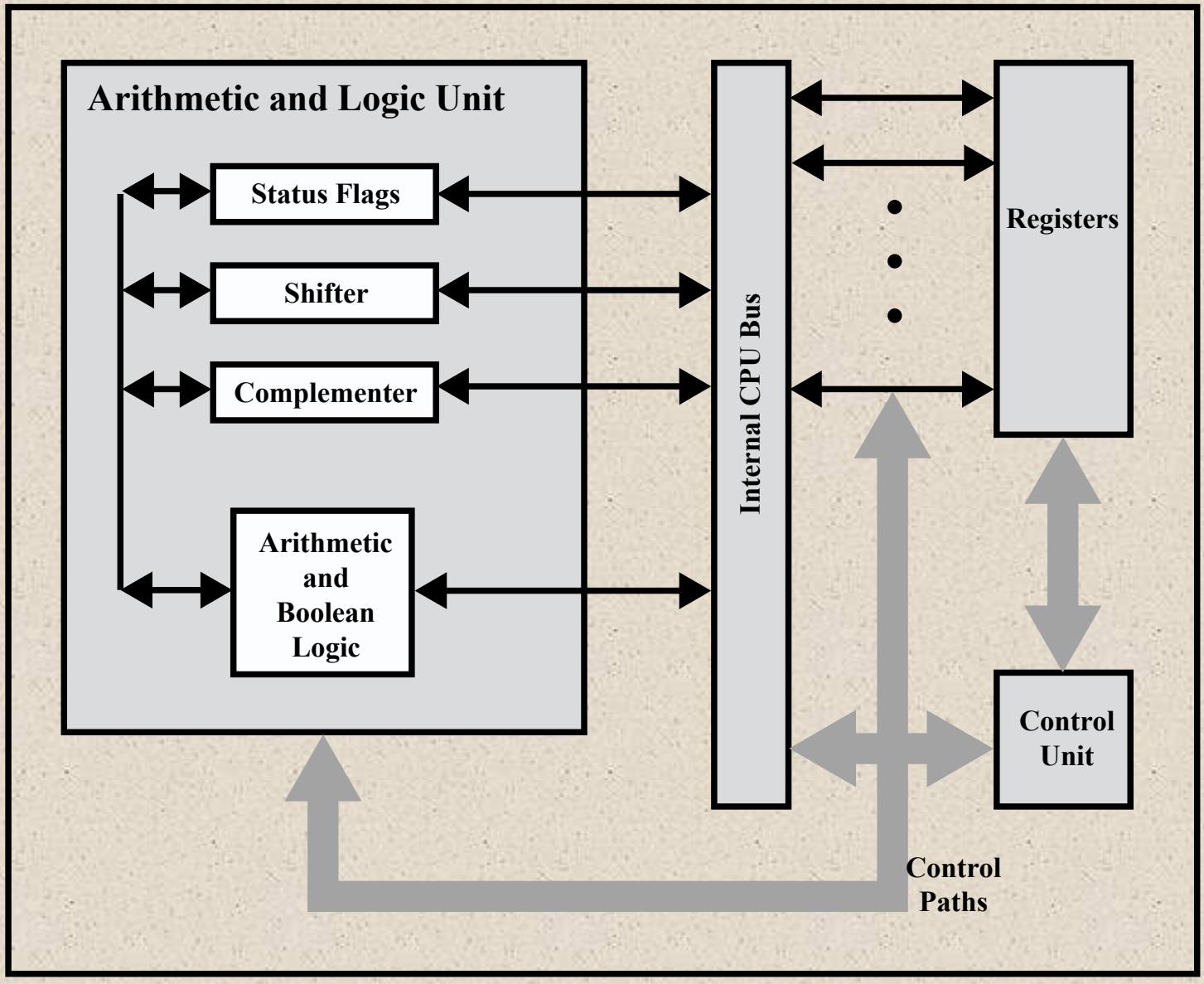


**Figure 14.2 Internal Structure of the CPU**

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# Register Organization

- Within the processor there is a set of registers that function as a level of memory above main memory and cache in the hierarchy
- The registers in the processor perform two roles:

**User-Visible Registers**

- Enable the machine or assembly language programmer to minimize main memory references by optimizing use of registers

**Control and Status Registers**

- Used by the control unit to control the operation of the processor and by privileged operating system programs to control the execution of programs

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# User-Visible Registers

Referenced by means of the machine language that the processor executes

## Categories:

- **General purpose**
  - Can be assigned to a variety of functions by the programmer
- **Data**
  - May be used only to hold data and cannot be employed in the calculation of an operand address
- **Address**
  - May be somewhat general purpose or may be devoted to a particular addressing mode
  - Examples: segment pointers, index registers, stack pointer
- **Condition codes**
  - Also referred to as *flags*
  - Bits set by the processor hardware as the result of operations

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# Table 14.1
# Condition Codes

| Advantages | Disadvantages |
|---|---|
| **1.** Because condition codes are set by normal arithmetic and data movement instructions, they should reduce the number of COMPARE and TEST instructions needed. | **1.** Condition codes add complexity, both to the hardware and software. Condition code bits are often modified in different ways by different instructions, making life more difficult for both the microprogrammer and compiler writer. |
| **2.** Conditional instructions, such as BRANCH are simplified relative to composite instructions, such as TEST AND BRANCH. | **2.** Condition codes are irregular; they are typically not part of the main data path, so they require extra hardware connections. |
| **3.** Condition codes facilitate multiway branches. For example, a TEST instruction can be followed by two branches, one on less than or equal to zero and one on greater than zero. | **3.** Often condition code machines must add special non-condition-code instructions for special situations anyway, such as bit checking, loop control, and atomic semaphore operations. |
| **4.** Condition codes can be saved on the stack during subroutine calls along with other register information. | **4.** In a pipelined implementation, condition codes require special synchronization to avoid conflicts. |

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# Control and Status Registers

Four registers are essential to instruction execution:

- Program counter (PC)
  - Contains the address of an instruction to be fetched
- Instruction register (IR)
  - Contains the instruction most recently fetched
- Memory address register (MAR)
  - Contains the address of a location in memory
- Memory buffer register (MBR)
  - Contains a word of data to be written to memory or the word most recently read

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# Program Status Word (PSW)

Register or set of registers that contain status information

Common fields or flags include:

- Sign
- Zero
- Carry
- Equal
- Overflow
- Interrupt Enable/Disable
- Supervisor

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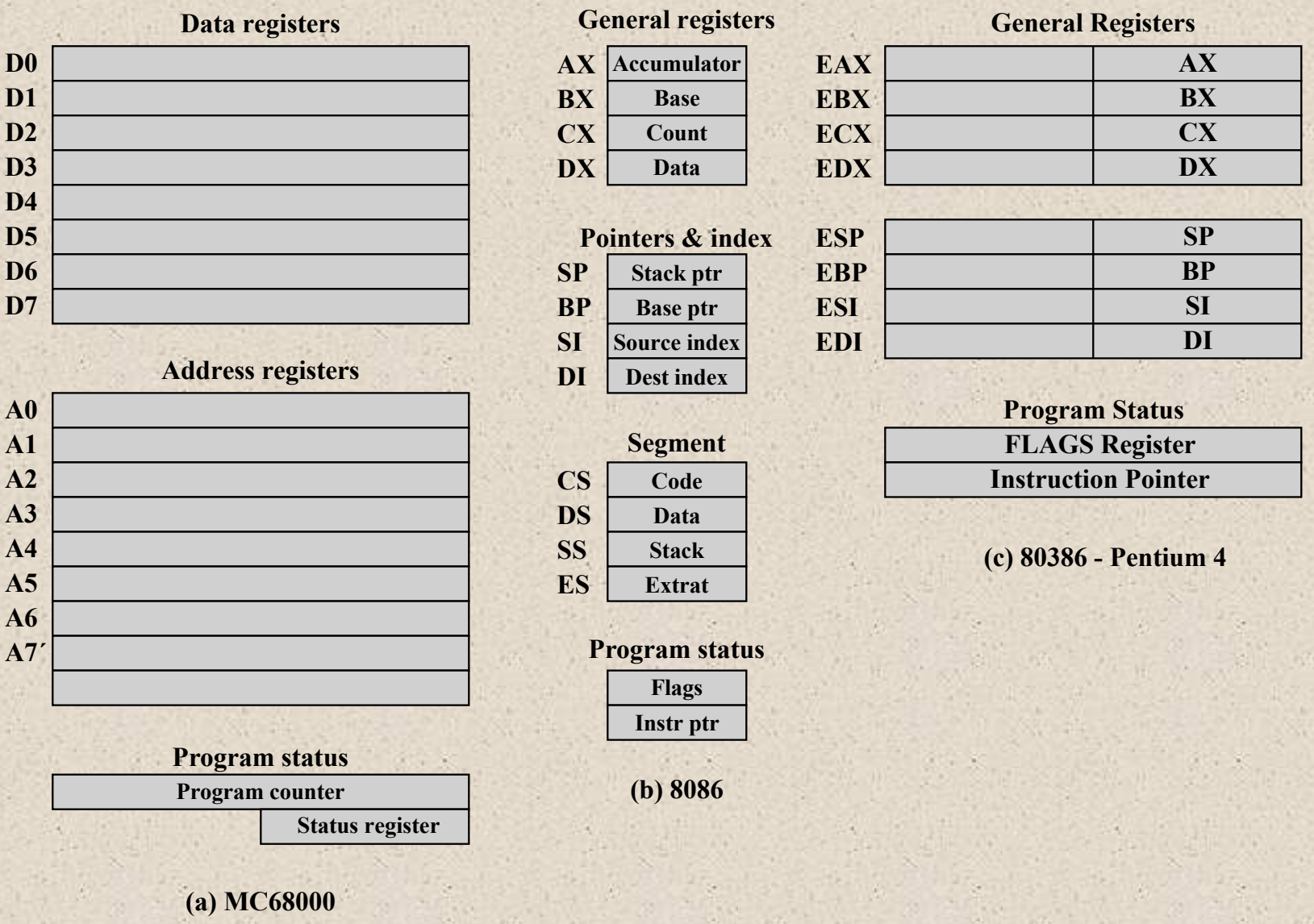


**Figure 14.3 Example Microprocessor Register Organizations**

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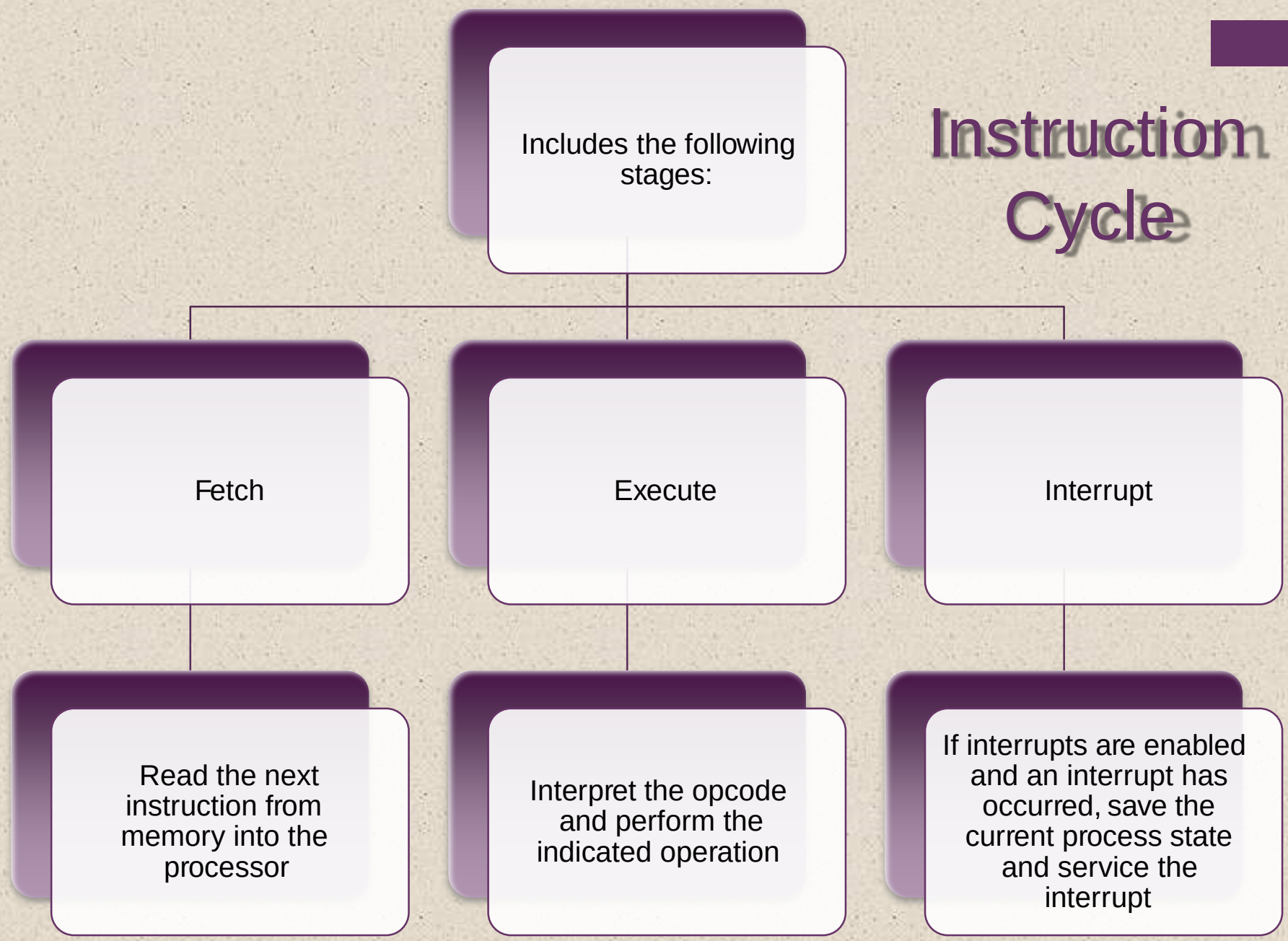
# Instruction Cycle

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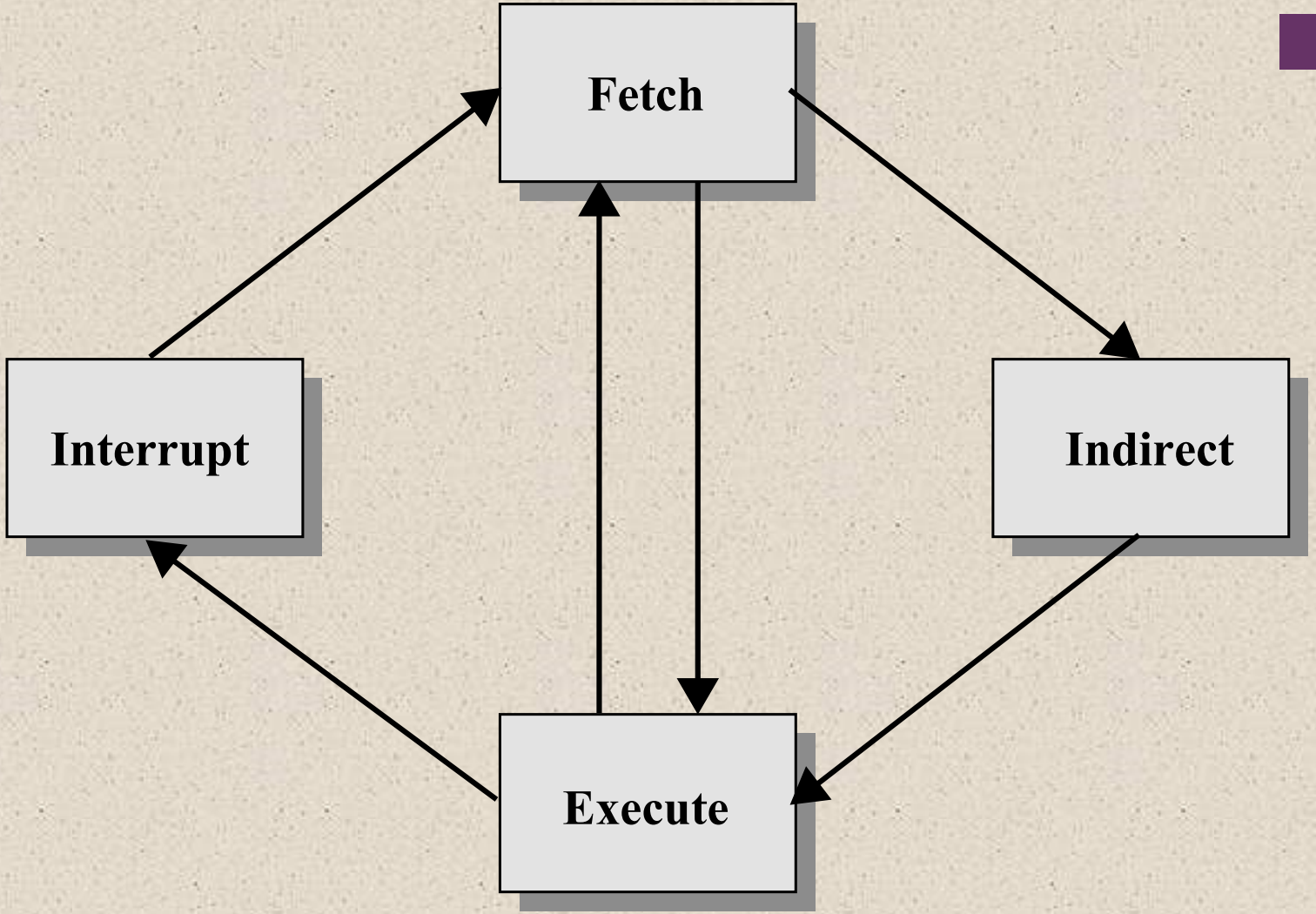


**Figure 14.4 The Instruction Cycle**

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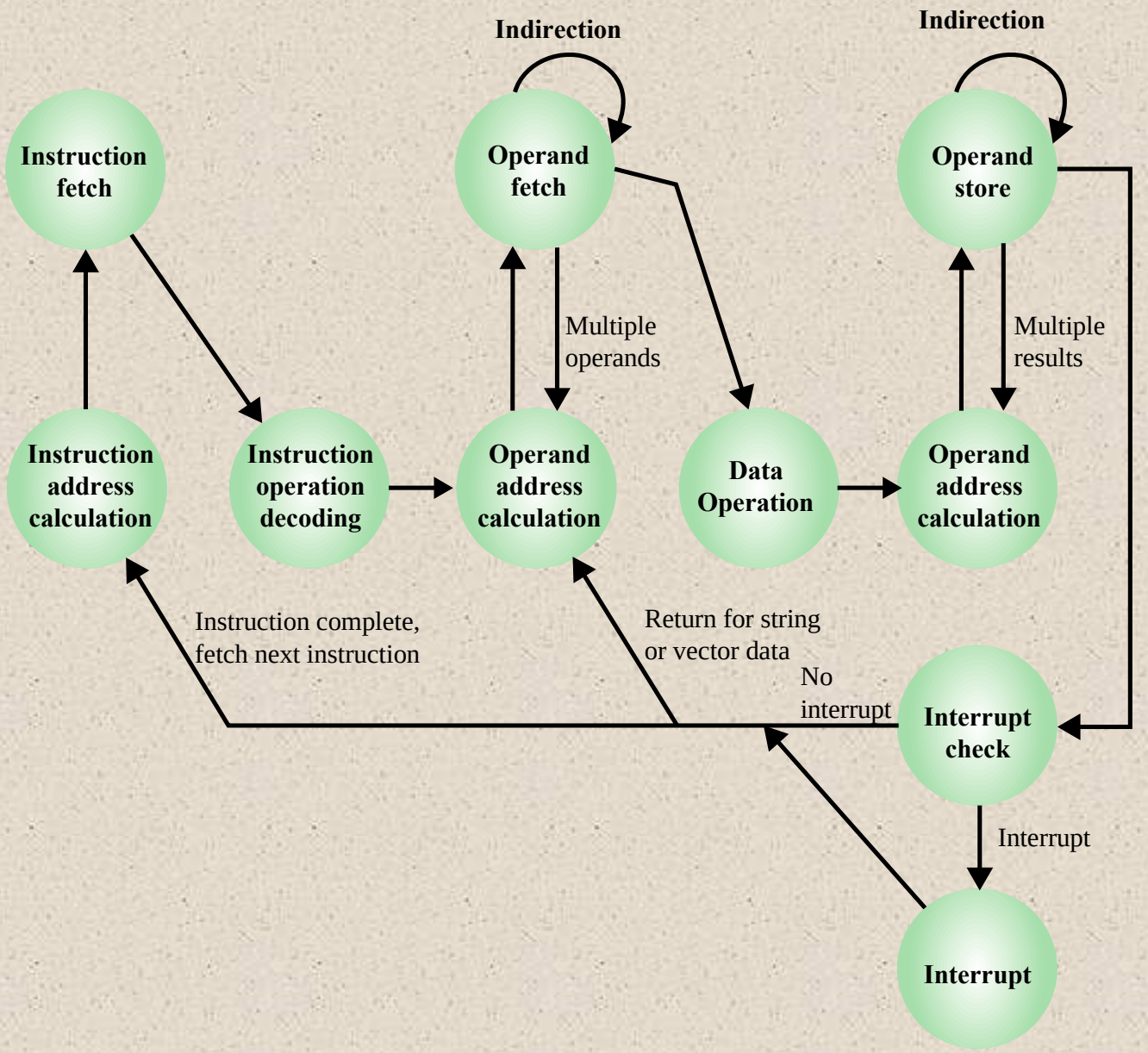


**Figure 14.5 Instruction Cycle State Diagram**

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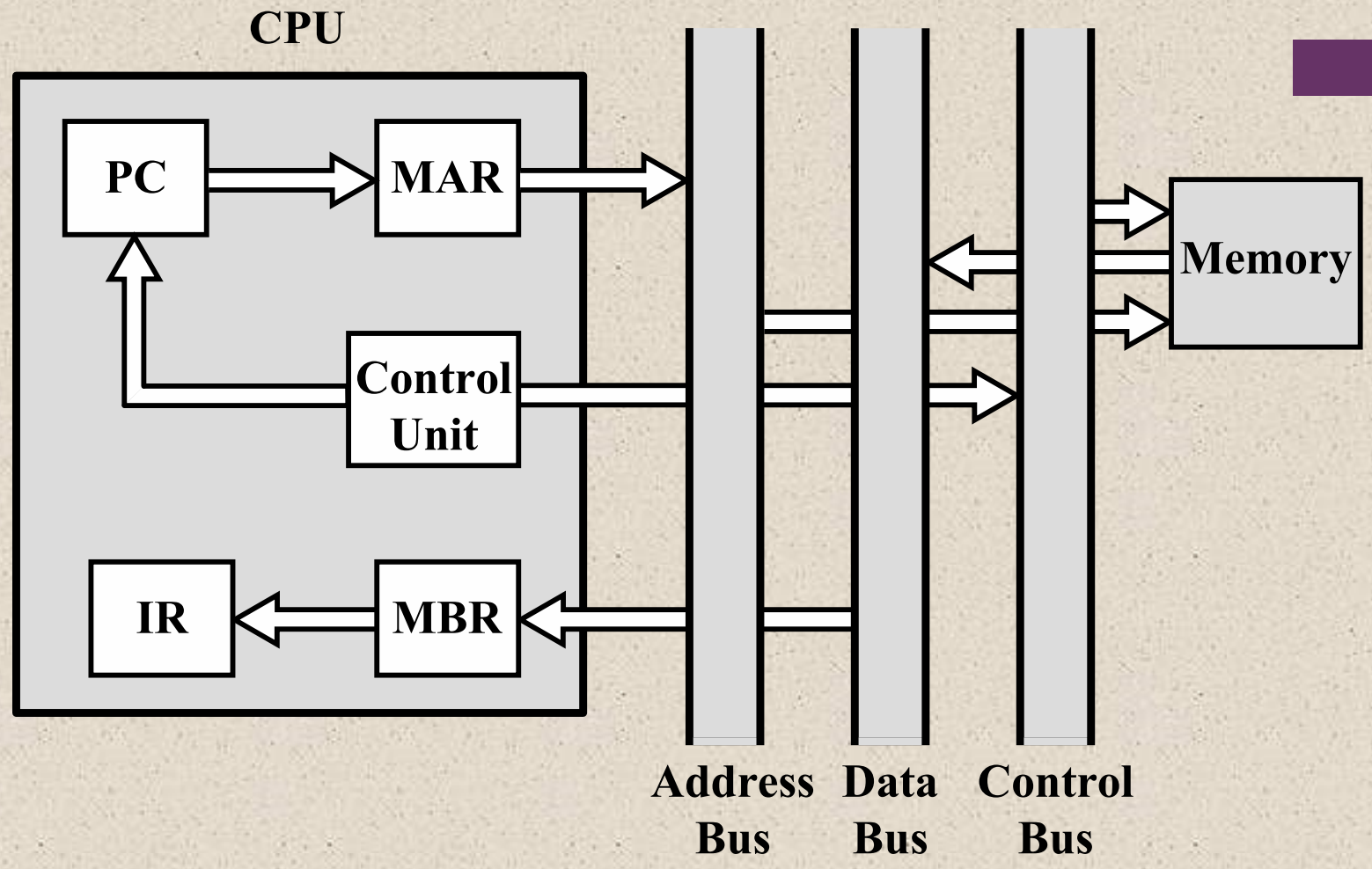


MBR = Memory buffer register
MAR = Memory address register
IR = Instruction register
PC = Program counter

**Figure 14.6 Data Flow, Fetch Cycle**

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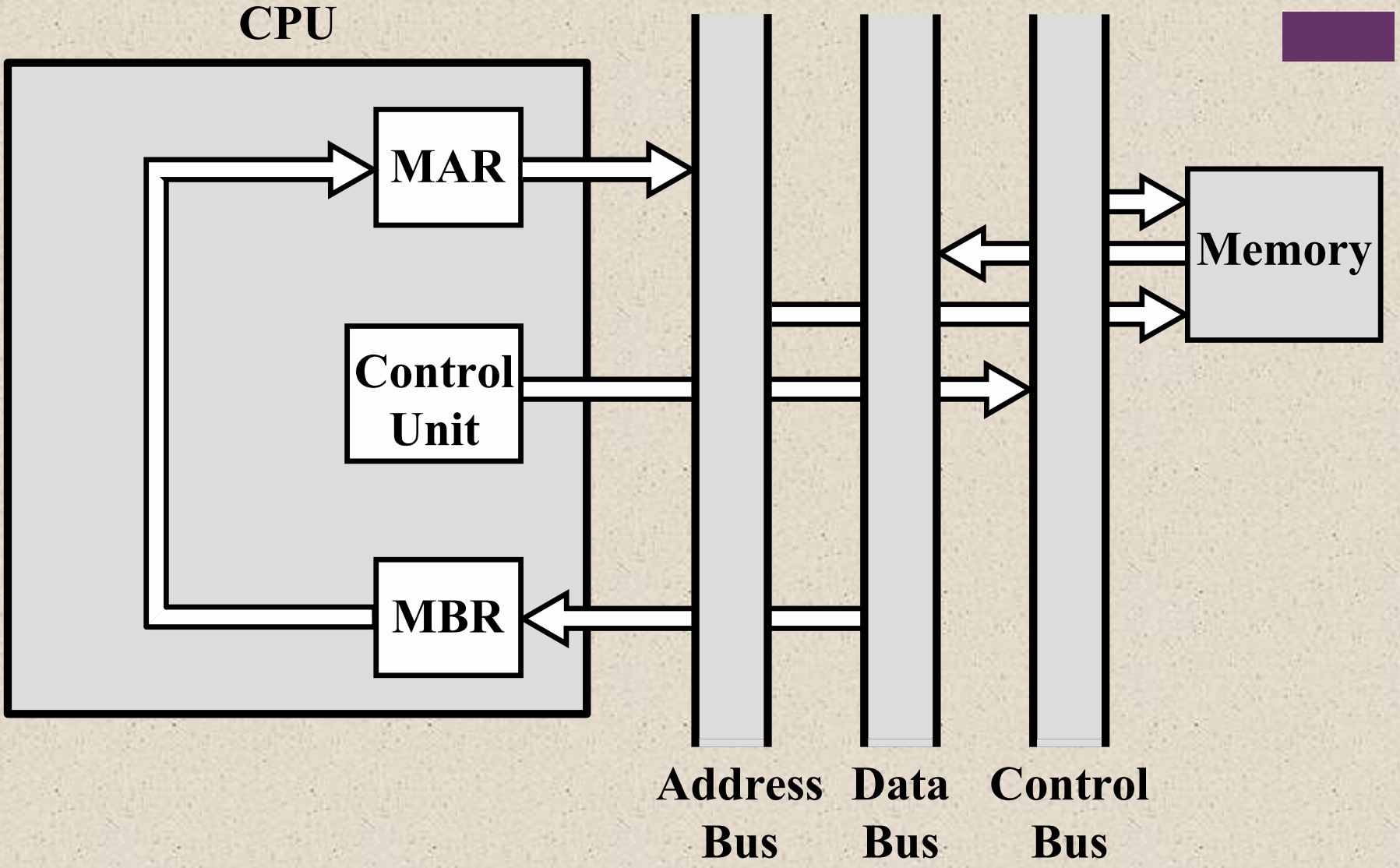


**Figure 14.7 Data Flow, Indirect Cycle**

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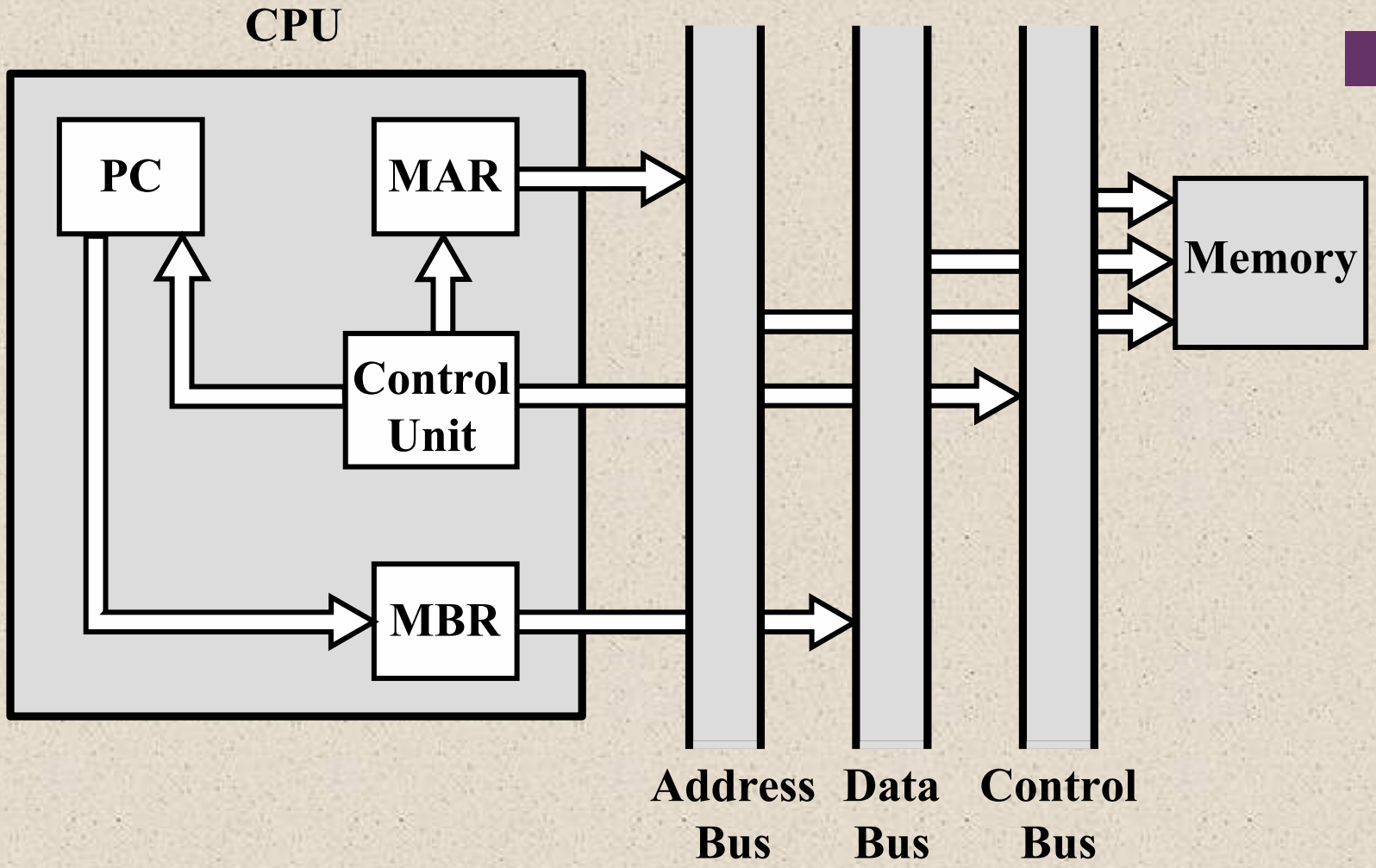


**Figure 14.8 Data Flow, Interrupt Cycle**

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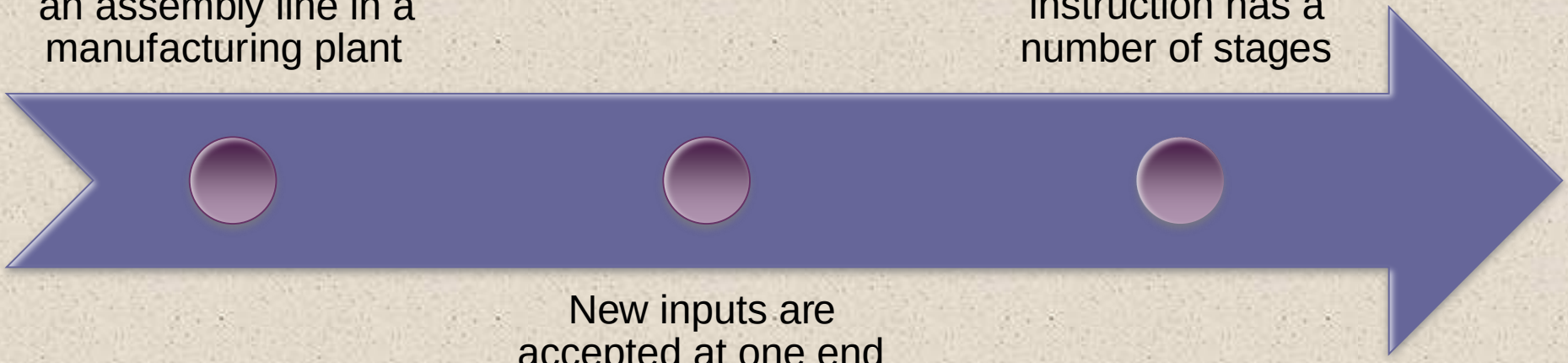
# Pipelining Strategy

Similar to the use of an assembly line in a manufacturing plant

To apply this concept to instruction execution we must recognize that an instruction has a number of stages



New inputs are accepted at one end before previously accepted inputs appear as outputs at the other end

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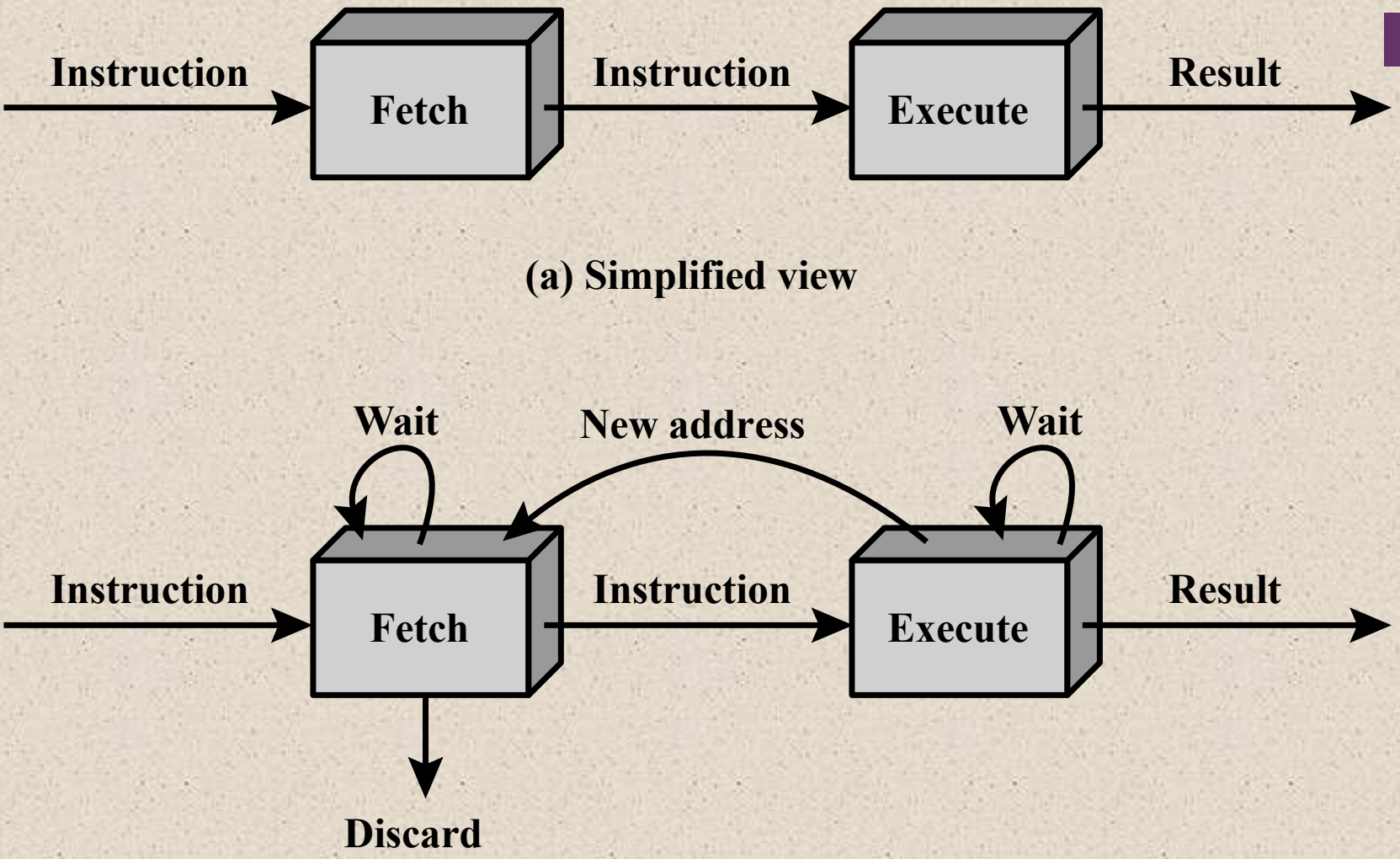


**Figure 14.9 Two-Stage Instruction Pipeline**

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# Additional Stages

- Fetch instruction (FI)
  - Read the next expected instruction into a buffer
- Decode instruction (DI)
  - Determine the opcode and the operand specifiers
- Calculate operands (CO)
  - Calculate the effective address of each source operand
  - This may involve displacement, register indirect, indirect, or other forms of address calculation
- Fetch operands (FO)
  - Fetch each operand from memory
  - Operands in registers need not be fetched
- Execute instruction (EI)
  - Perform the indicated operation and store the result, if any, in the specified destination operand location
- Write operand (WO)
  - Store the result in memory

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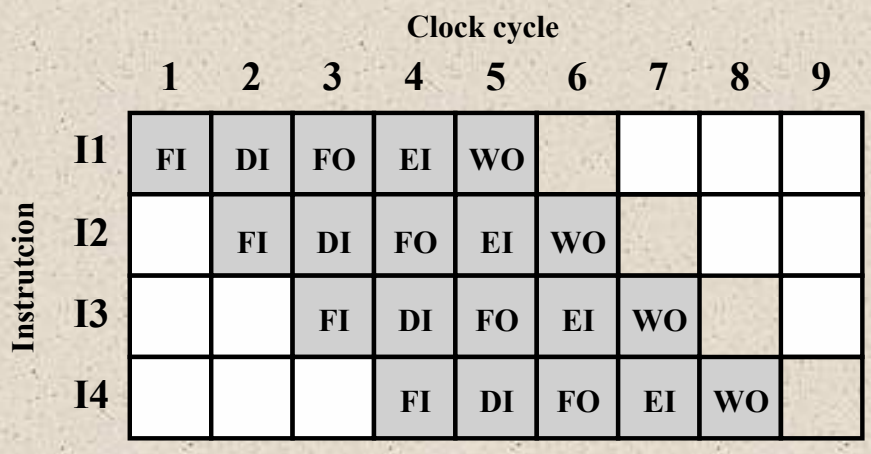
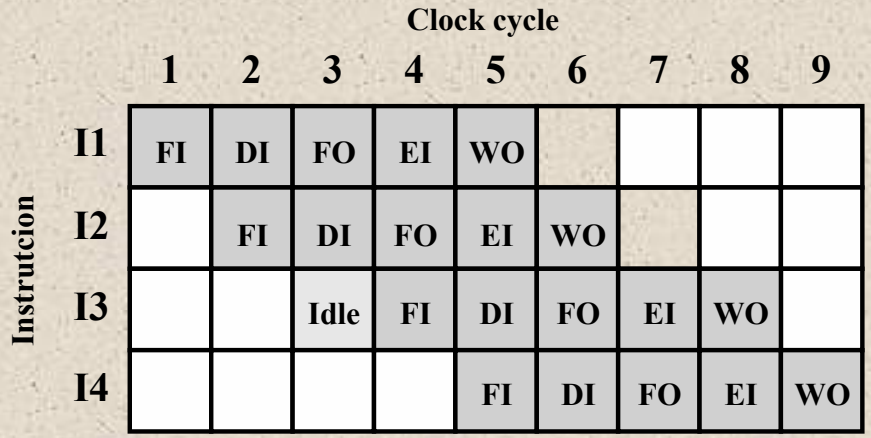



(a) Five-stage pipeline, ideal case




(b) I1 source operand in memory

**Figure 14.15 Example of Resource Hazard**

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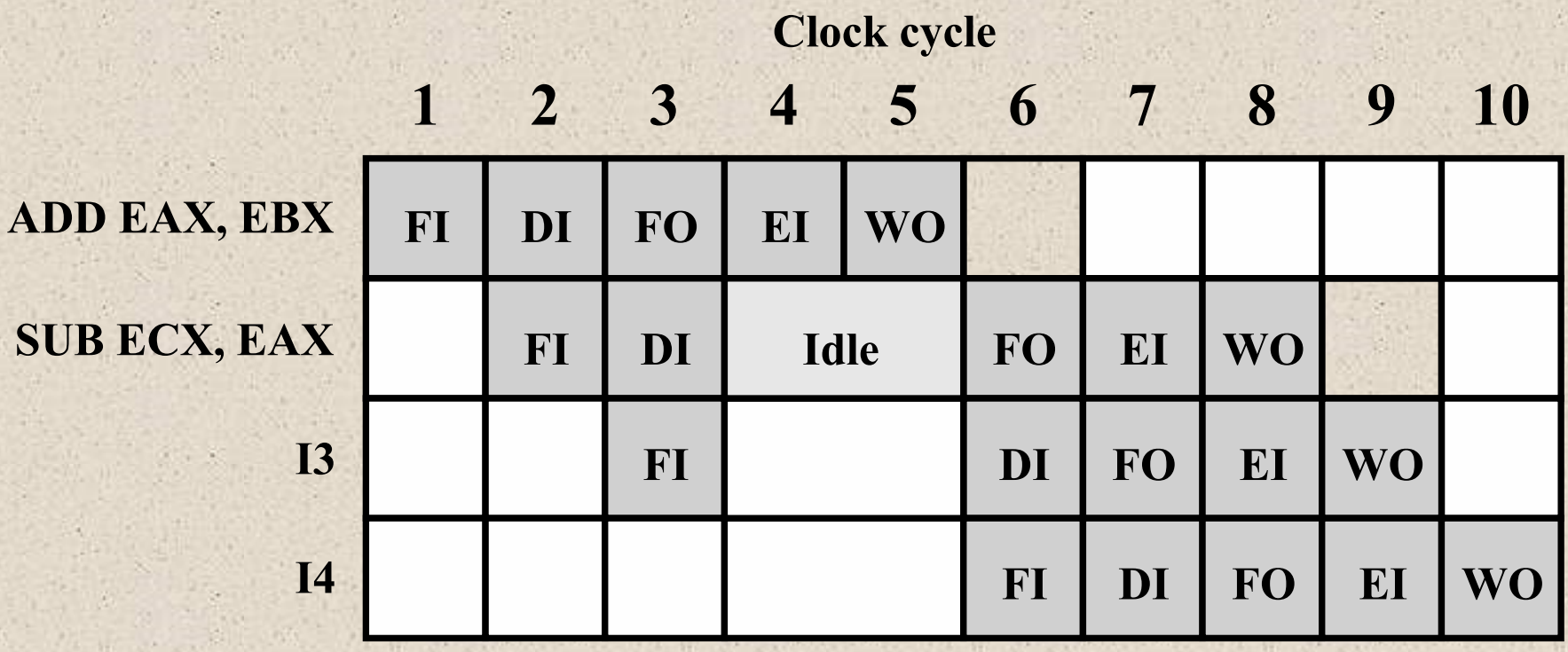


**Figure 14.16 Example of Data Hazard**

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# Types of Data Hazard

- Read after write (RAW), or true dependency
  - An instruction modifies a register or memory location
  - Succeeding instruction reads data in memory or register location
  - Hazard occurs if the read takes place before write operation is complete
- Write after read (WAR), or antidependency
  - An instruction reads a register or memory location
  - Succeeding instruction writes to the location
  - Hazard occurs if the write operation completes before the read operation takes place
- Write after write (WAW), or output dependency
  - Two instructions both write to the same location
  - Hazard occurs if the write operations take place in the reverse order of the intended sequence

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# Control Hazard

- Also known as a *branch hazard*
- Occurs when the pipeline makes the wrong decision on a branch prediction
- Brings instructions into the pipeline that must subsequently be discarded
- Dealing with Branches:
  - Multiple streams
  - Prefetch branch target
  - Loop buffer
  - Branch prediction
  - Delayed branch

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# Multiple Streams

A simple pipeline suffers a penalty for a branch instruction because it must choose one of two instructions to fetch next and may make the wrong choice

A brute -force approach is to replicate the initial portions of the pipeline and allow the pipeline to fetch both instructions, making use of two streams

Drawbacks:

- With multiple pipelines there are contention delays for access to the registers and to memory
- Additional branch instructions may enter the pipeline before the original branch decision is resolved

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# Loop Buffer

- Small, very -high speed memory maintained by the instruction fetch stage of the pipeline and containing the $n$ most recently fetched instructions, in sequence
- Benefits:
  - Instructions fetched in sequence will be available without the usual memory access time
  - If a branch occurs to a target just a few locations ahead of the address of the branch instruction, the target will already be in the buffer
  - This strategy is particularly well suited to dealing with loops
- Similar in principle to a cache dedicated to instructions
  - Differences:
    - The loop buffer only retains instructions in sequence
    - Is much smaller in size and hence lower in cost

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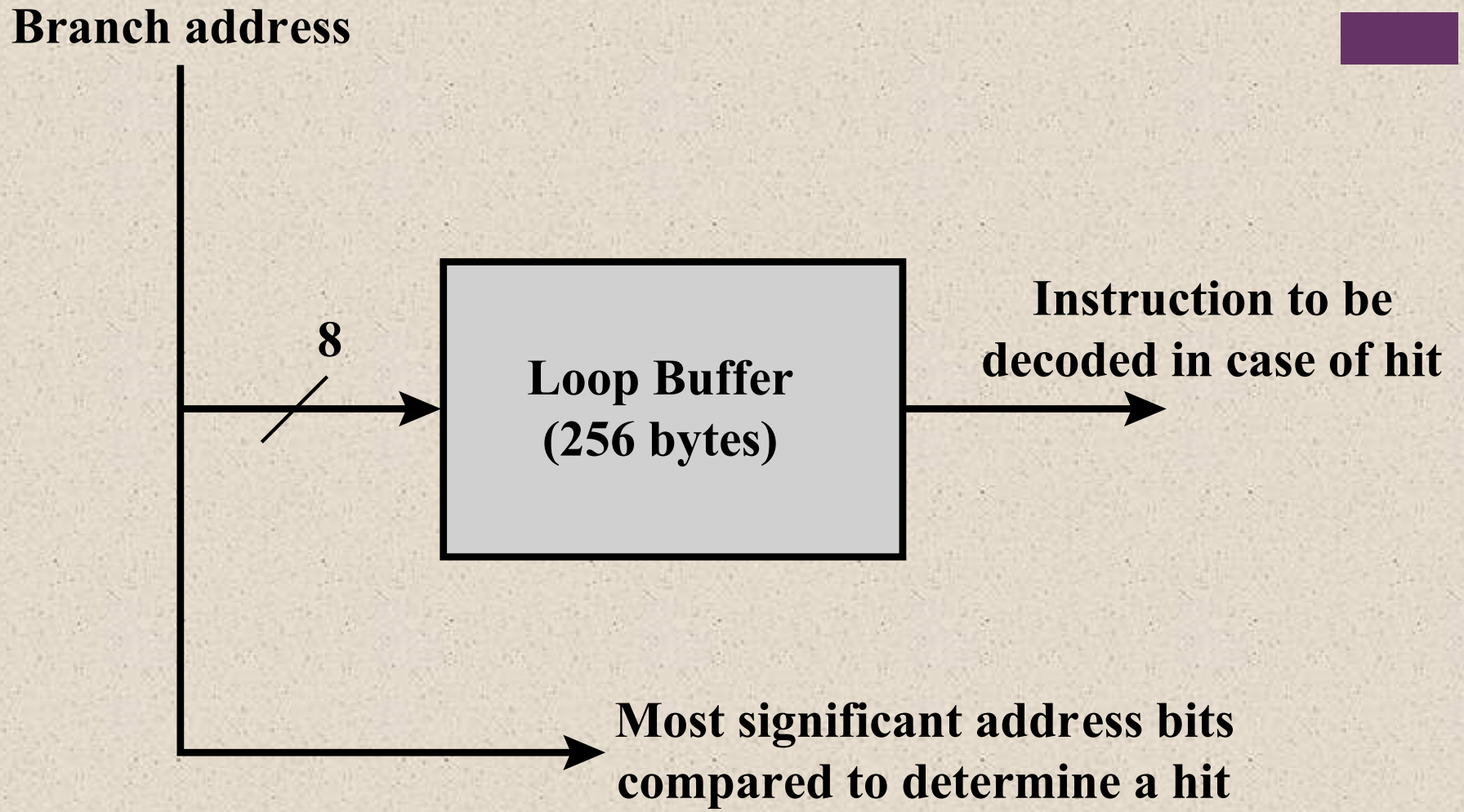


**Figure 14.17 Loop Buffer**

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# Branch Prediction

- Various techniques can be used to predict whether a branch will be taken:

1. Predict never taken
2. Predict always taken
3. Predict by opcode

- These approaches are static
- They do not depend on the execution history up to the time of the conditional branch instruction

1. Taken/not taken switch
2. Branch history table

- These approaches are dynamic
- They depend on the execution history

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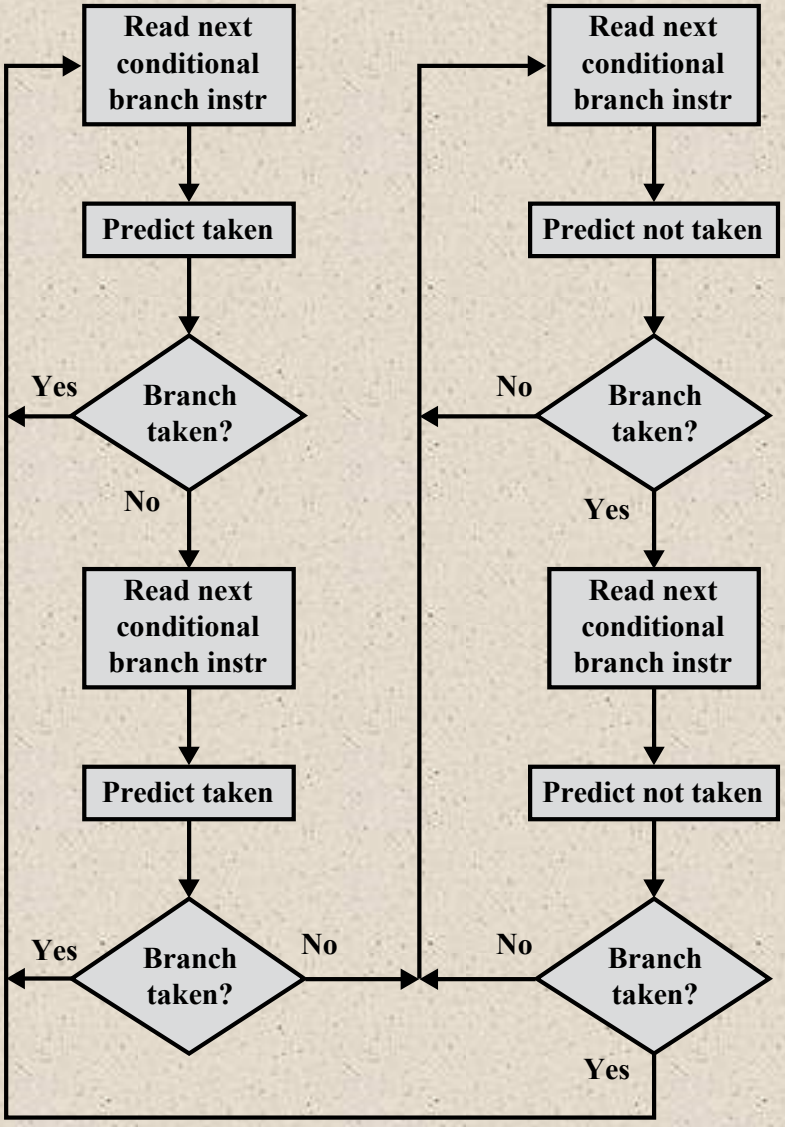


**Figure 14.18 Branch Prediction Flow Chart**

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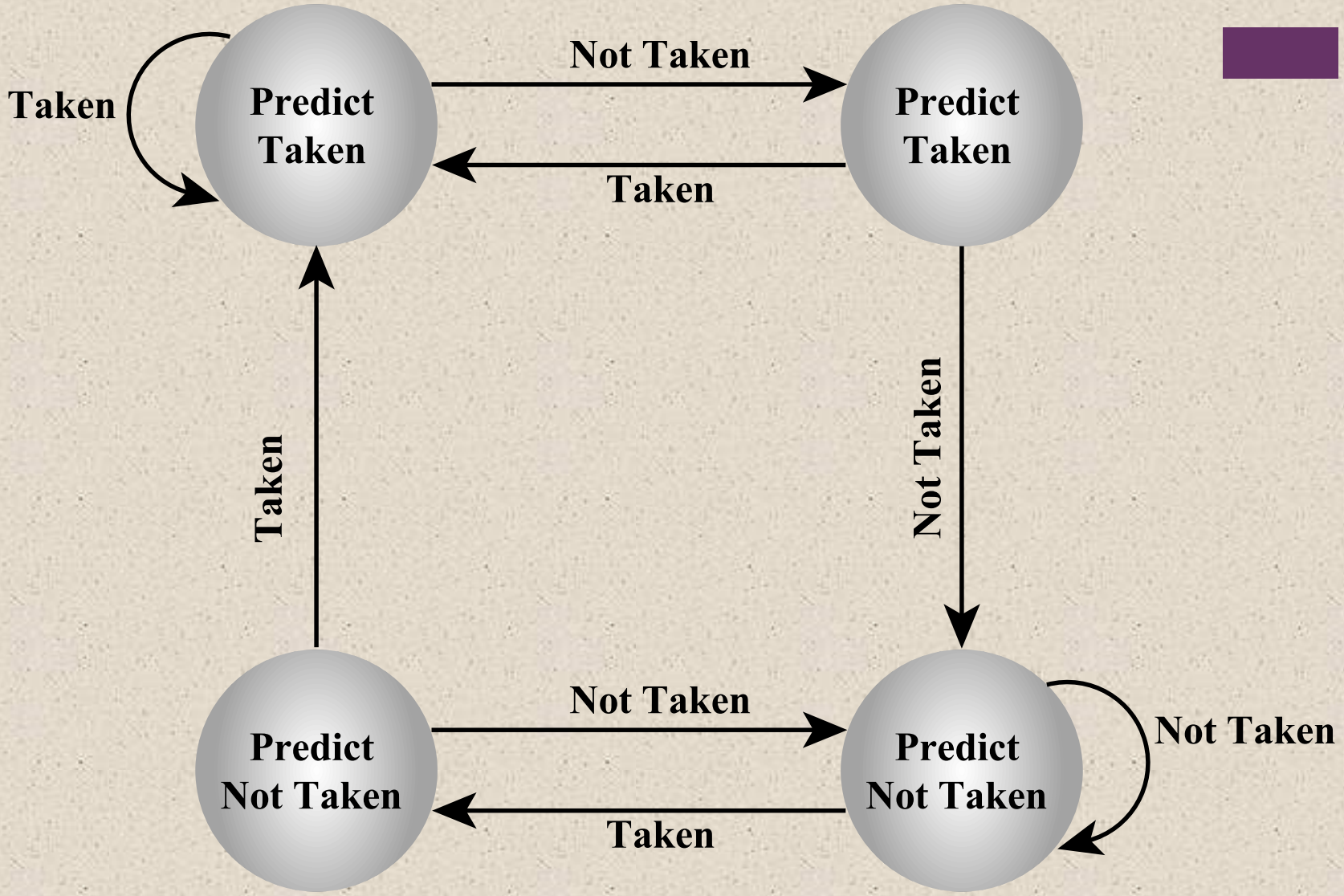


**Figure 14.19 Branch Prediction State Diagram**

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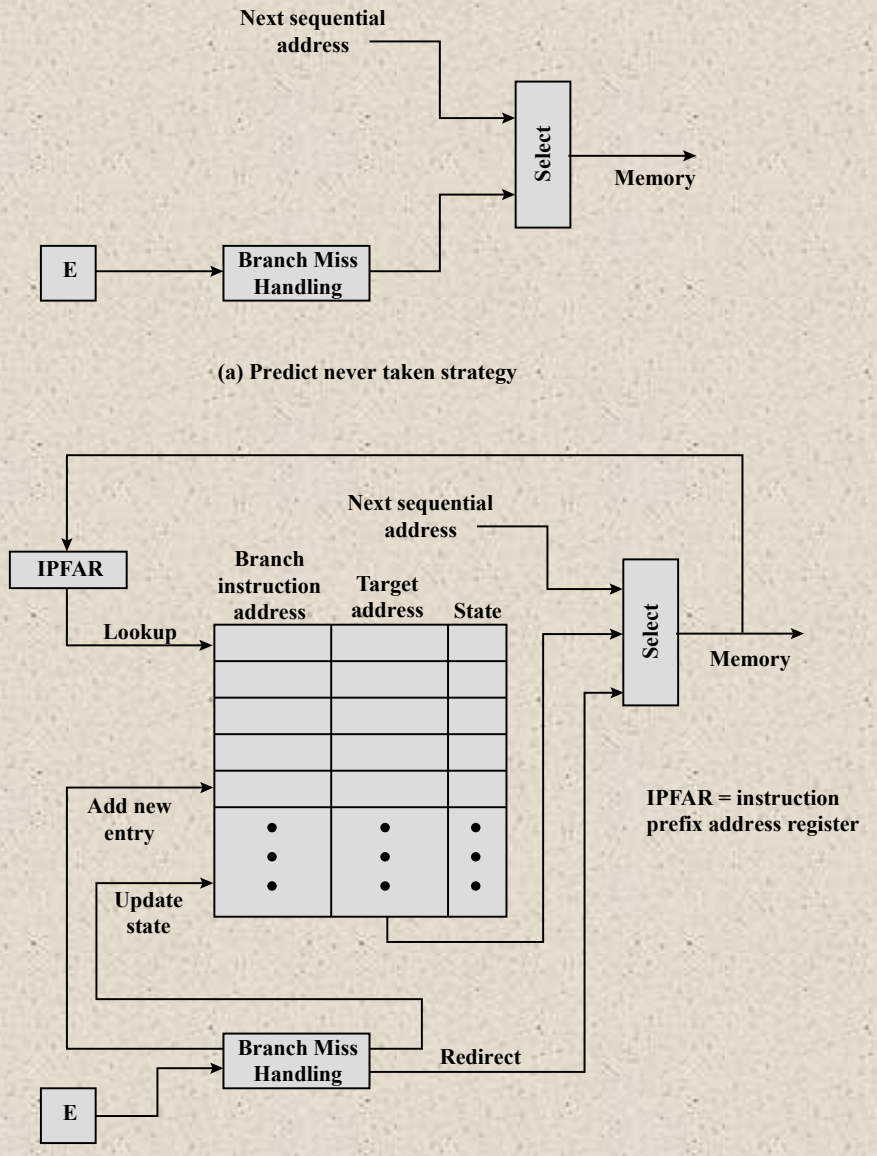


**Figure 14.20 Dealing with Branches**

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# Intel 80486 Pipelining

## Fetch

- Objective is to fill the prefetch buffers with new data as soon as the old data have been consumed by the instruction decoder
- Operates independently of the other stages to keep the prefetch buffers full

## Decode stage 1

- All opcode and addressing -mode information is decoded in the D1 stage
- 3 bytes of instruction are passed to the D1 stage from the prefetch buffers
- D1 decoder can then direct the D2 stage to capture the rest of the instruction

## Decode stage 2

- Expands each opcode into control signals for the ALU
- Also controls the computation of the more complex addressing modes

## Execute

Stage includes ALU operations, cache access, and register update

## Write back

Updates registers and status flags modified during the preceding execute stage

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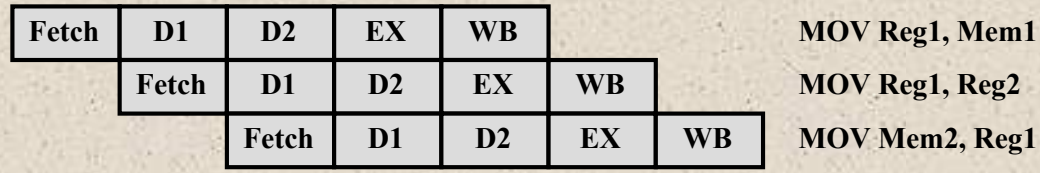


(a) No Data Load Delay in the Pipeline

(b) Pointer Load Delay

(c) Branch Instruction Timing

**Figure 14.21 80486 Instruction Pipeline Examples**

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# Table 14.2

# x86 Processor Registers

**(a) Integer Unit in 32-bit Mode**

| Type | Number | Length (bits) | Purpose |
|---|---|---|---|
| General | 8 | 32 | General-purpose user registers |
| Segment | 6 | 16 | Contain segment selectors |
| EFLAGS | 1 | 32 | Status and control bits |
| Instruction Pointer | 1 | 32 | Instruction pointer |

**(b) Integer Unit in 64-bit Mode**

| Type | Number | Length (bits) | Purpose |
|---|---|---|---|
| General | 16 | 32 | General-purpose user registers |
| Segment | 6 | 16 | Contain segment selectors |
| RFLAGS | 1 | 64 | Status and control bits |
| Instruction Pointer | 1 | 64 | Instruction pointer |

**(c) Floating-Point Unit**

| Type | Number | Length (bits) | Purpose |
|---|---|---|---|
| Numeric | 8 | 80 | Hold floating-point numbers |
| Control | 1 | 16 | Control bits |
| Status | 1 | 16 | Status bits |
| Tag Word | 1 | 16 | Specifies contents of numeric registers |
| Instruction Pointer | 1 | 48 | Points to instruction interrupted by exception |
| Data Pointer | 1 | 48 | Points to operand interrupted by exception |

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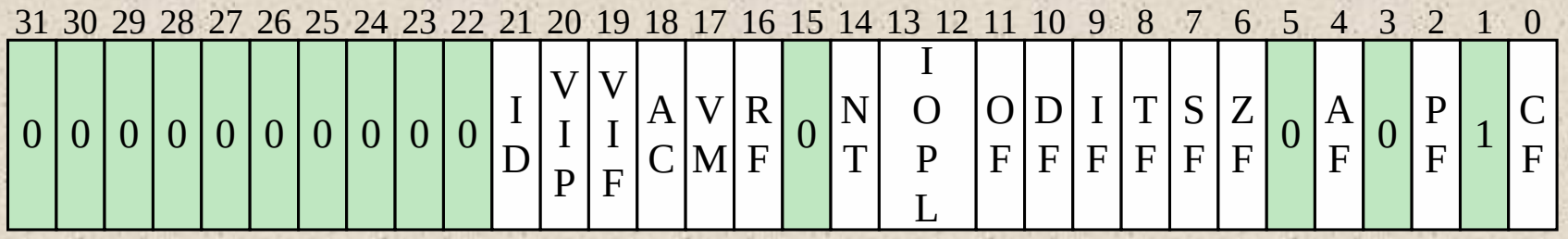


X ID = Identification flag
X VIP = Virtual interrupt pending
X VIF = Virtual interrupt flag
X AC = Alignment check
X VM = Virtual 8086 mode
X RF = Resume flag
X NT = Nested task flag
X IOPL = I/O privilege level
S OF = Overflow flag

C DF = Direction flag
X IF = Interrupt enable flag
X TF = Trap flag
S SF = Sign flag
S ZF = Zero flag
S AF = Auxiliary carry flag
S PF = Parity flag
S CF = Carry flag

S Indicates a Status Flag
C Indicates a Control Flag
X Indicates a System Flag
Shaded bits are reserved

**Figure 14.22 x86 EFLAGS Register**

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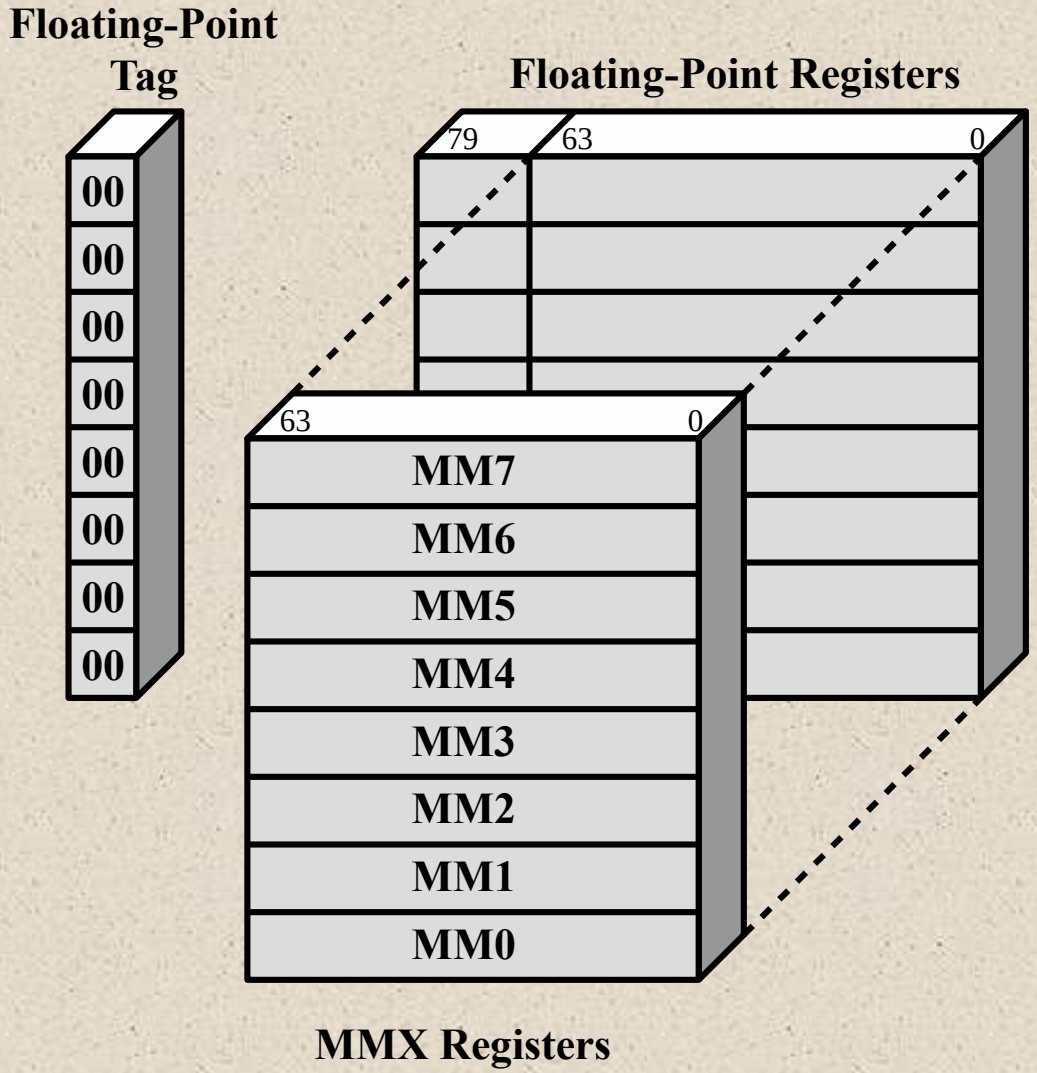


**Figure 14.24 Mapping of MMX Registers to Floating-Point Registers**

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# Interrupt Processing

## Interrupts and Exceptions

- Interrupts
  - Generated by a signal from hardware and it may occur at random times during the execution of a program
  - Maskable
  - Nonmaskable
- Exceptions
  - Generated from software and is provoked by the execution of an instruction
  - Processor detected
  - Programmed
- Interrupt vector table
  - Every type of interrupt is assigned a number
  - Number is used to index into the interrupt vector table

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Unshaded: 2xceptions

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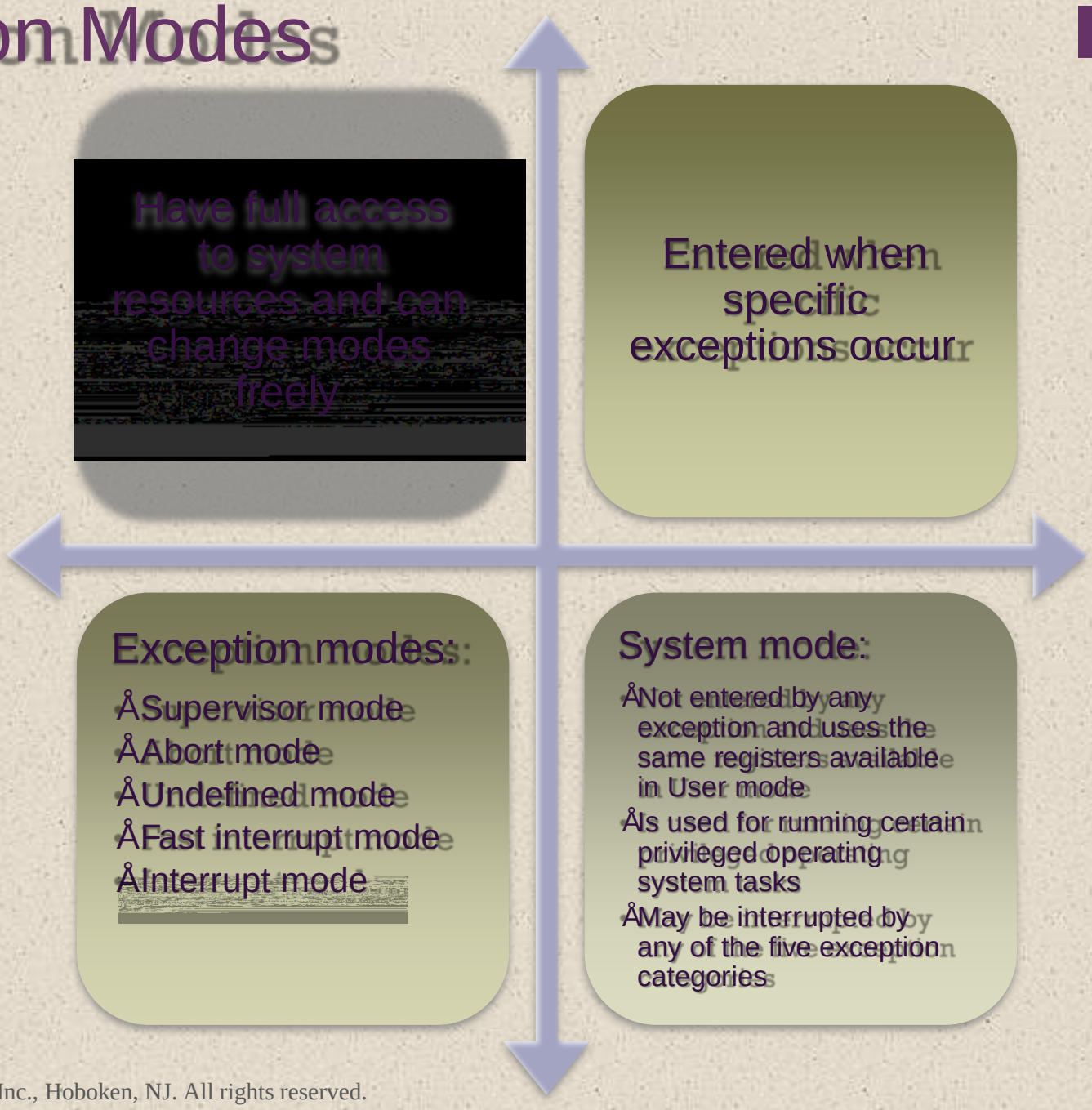
# Exception Modes

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| Modes | | | | | | |
|---|---|---|---|---|---|---|
| | Privileged modes | | | | | |
| | | Exception modes | | | | |
| **User** | **System** | **Supervisor** | **Abort** | **Undefined** | **Interrupt** | **Fast Interrupt** |
| R0 | R0 | R0 | R0 | R0 | R0 | R0 |
| R1 | R1 | R1 | R1 | R1 | R1 | R1 |
| R2 | R2 | R2 | R2 | R2 | R2 | R2 |
| R3 | R3 | R3 | R3 | R3 | R3 | R3 |
| R4 | R4 | R4 | R4 | R4 | R4 | R4 |
| R5 | R5 | R5 | R5 | R5 | R5 | R5 |
| R6 | R6 | R6 | R6 | R6 | R6 | R6 |
| R7 | R7 | R7 | R7 | R7 | R7 | R7 |
| R8 | R8 | R8 | R8 | R8 | R8 | R8_fiq |
| R9 | R9 | R9 | R9 | R9 | R9 | R9_fiq |
| R10 | R10 | R10 | R10 | R10 | R10 | R10_fiq |
| R11 | R11 | R11 | R11 | R11 | R11 | R11_fiq |
| R12 | R12 | R12 | R12 | R12 | R12 | R12_fiq |
| R13 (SP) | R13 (SP) | R13_svc | R13_abt | R13_und | R13_irq | R13_fiq |
| R14 (LR) | R14 (LR) | R14_svc | R14_abt | R14_und | R14_irq | R14_fiq |
| R15 (PC) | R15 (PC) | R15 (PC) | R15 (PC) | R15 (PC) | R15 (PC) | R15 (PC) |
| CPSR | CPSR | CPSR | CPSR | CPSR | CPSR | CPSR |
| | | SPSR_svc | SPSR_abt | SPSR_und | SPSR_irq | SPSR_fiq |

Shading indicates that the normal register used by User or System mode has been replaced by an alternative register specific to the exception mode.

SP = stack pointer
LR = link register
PC = program counter

CPSR = current program status register
SPSR = saved program status register

**Figure 14.26 ARM Register Organization**

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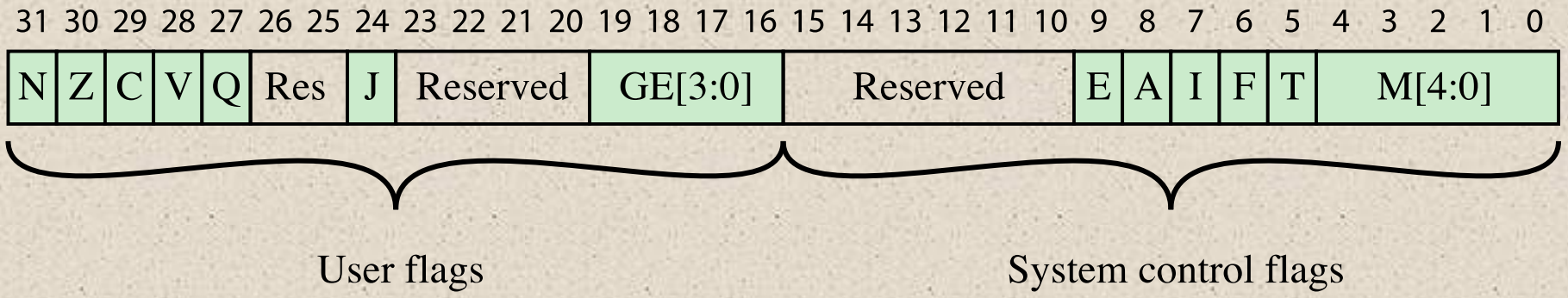


**Figure 14.27 Format of ARM CPSR AND SPSR**

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| Exception type | Mode | Normal entry address | Description |
|---|---|---|---|
| Reset | Supervisor | 0x00000000 | Occurs when the system is initialized. |
| Data abort | Abort | 0x00000010 | Occurs when an invalid memory address has been accessed, such as if there is no physical memory for an address or the correct access permission is lacking. |
| FIQ (fast interrupt) | FIQ | 0x0000001C | Occurs when an external device asserts the FIQ pin on the processor. An interrupt cannot be interrupted except by an FIQ. FIQ is designed to support a data transfer or channel process, and has sufficient private registers to remove the need for register saving in such applications, therefore minimizing the overhead of context switching. A fast interrupt cannot be interrupted. |
| IRQ (interrupt) | IRQ | 0x00000018 | Occurs when an external device asserts the IRQ pin on the processor. An interrupt cannot be interrupted except by an FIQ. |
| Prefetch abort | Abort | 0x0000000C | Occurs when an attempt to fetch an instruction results in a memory fault. The exception is raised when the instruction enters the execute stage of the pipeline. |
| Undefined instructions | Undefined | 0x00000004 | Occurs when an instruction not in the instruction set reaches the execute stage of the pipeline. |
| Software interrupt | Supervisor | 0x00000008 | Generally used to allow user mode programs to call the OS. The user program executes a SWI instruction with an argument that identifies the function the user wishes to perform. |

# Table 14.4

## ARM Interrupt Vector

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# Summary

## Chapter 14

## Processor Structure and Function

- Processor organization
- Register organization
  - User-visible registers
  - Control and status registers
- Instruction cycle
  - The indirect cycle
  - Data flow
- The x86 processor family
  - Register organization
  - Interrupt processing
- Instruction pipelining
  - Pipelining strategy
  - Pipeline performance
  - Pipeline hazards
  - Dealing with branches
  - Intel 80486 pipelining
- The Arm processor
  - Processor organization
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  - Register organization
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